Roll No. | | | | | | | | | | |

Total No. of Pages : 02

Total No. of Questions : 13

B.Pharmacy (Sem.-8)

ADVANCED INSTRUMENTATION TECHNIQUES

Subject Code : BP811ET

M.Code : 79774

Date of Examination : 23-05-2024

Time : 3 Hrs.

Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.
3. SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.

SECTION-A

1. Write briefly :

(a) Carbon-13 is detected in NMR but Carbon-12 is not. Why?

(b) What is Nitrogen rule? Give its significance.

(c) Describe Brag's law.

(d) Enumerate different types of crystals.

(e) What is system suitability is validation?

(f) Give any two elements used in Radioimmuno assays with their specific applications.

(g) What is the principle of solid-phase extraction?

(h) Differentiate between Benzyl acetate and Phenyl propionate on the basis of NMR spectra.

(i) What is tropylium ion? Give its significance in structural elucidation.

(j) Give limitations of GC-MS over LC-MS.

SECTION-B

2. Write a comparative account of DSC, DTA and TGA. Give their specific applications.

3. What is the principle of LC-MS/MS? Describe the working of a typical LC-MS instrument with the help of a block diagram. Write a detailed account on various analyzers used in it.

4. Describe various methods of ionization used in mass spectrometry.

SECTION-C

5. What is spin-spin splitting? Explain the mechanism of splitting of signal with examples.

6. Define the term magnetic anisotropy. How does it affect precessional frequency of protons?

7. Explain the working of TOF analyzer. Enumerate its advantages and limitations.

8. Describe the working of rotating crystal technique.

9. What is the importance method validation? Write an account on USFDA guidelines for method validation.

10. What is the principle of RIA? Describe its methodologies and applications.

11. Write a note on principle, methodology and applications of liquid-liquid extraction.

12. What are isotope peaks? Discuss their importance with specific examples.

13. Give a step-by-step procedure for calibration of fluorimeter.

NOTE : Disclosure of Identity by writing Mobile No. or Marking of passing request on any paper of Answer Sheet will lead to UMC against the Student.

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